

PROGRAM

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

// Array of 5 semaphores representing the 5 chopsticks
sem_t chopstick[5];

void *philosopher(void *arg) {
    int i = *(int *)arg;

    // Infinite loop as per the algorithm: do { ... } while(1);
    while(1) {
        printf("Philosopher %d is THINKING.\n", i);
        sleep(1); // Sleep to simulate thinking time

        printf("Philosopher %d is hungry and waiting for chopsticks.\n", i);

        // Pick up chopsticks
        sem_wait(&chopstick[i]); // wait(chopstick[i])
        sem_wait(&chopstick[(i+1) % 5]); // wait(chopstick[(i+1)%5])

        // Eating the rice
        printf("Philosopher %d is EATING THE RICE.\n", i);
        sleep(2); // Sleep to simulate eating time

        // Put down chopsticks
        sem_post(&chopstick[i]); // signal(chopstick[i])
        sem_post(&chopstick[(i+1) % 5]); // signal(chopstick[(i+1)%5])

        printf("Philosopher %d finished eating and put down chopsticks.\n", i);
    }
}

int main() {
    pthread_t tid[5];
    int phil_num[5];

    // Initialize the semaphores to 1
    for(int i = 0; i < 5; i++) {
        sem_init(&chopstick[i], 0, 1);
    }

    printf("Dining Philosophers Simulation Started (Press Ctrl+C to stop)...\n\n");

    // Create the philosopher threads
    for(int i = 0; i < 5; i++) {
        phil_num[i] = i;
        pthread_create(&tid[i], NULL, philosopher, &phil_num[i]);
    }

    // Wait for the threads to finish (Technically unreachable due to while(1))
    for(int i = 0; i < 5; i++) {
        pthread_join(tid[i], NULL);
    }
}
```

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}

// Destroy semaphores
for(int i = 0; i < 5; i++) {
    sem_destroy(&chopstick[i]);
}

return 0;
}
```

OUTPUT

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_6$ gcc 6.c
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_6$ ./a.out
Dining Philosophers Simulation Started (Press Ctrl+C to stop)...
```

```
Philosopher 0 is THINKING.
Philosopher 1 is THINKING.
Philosopher 2 is THINKING.
Philosopher 3 is THINKING.
Philosopher 4 is THINKING.
Philosopher 0 is hungry and waiting for chopsticks.
Philosopher 0 is EATING THE RICE.
Philosopher 1 is hungry and waiting for chopsticks.
Philosopher 3 is hungry and waiting for chopsticks.
Philosopher 2 is hungry and waiting for chopsticks.
Philosopher 4 is hungry and waiting for chopsticks.
Philosopher 3 is EATING THE RICE.
Philosopher 0 finished eating and put down chopsticks.
Philosopher 0 is THINKING.
Philosopher 4 is EATING THE RICE.
Philosopher 3 finished eating and put down chopsticks.
Philosopher 3 is THINKING.
Philosopher 2 is EATING THE RICE.
^C
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_6$
```